

GBBO Data Cleaning/Joining

v1

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Data Cleaning

I combine and clean several sources of data on the Netflix seasons of *The Great British Bake Off*, including:

- Judging data prepared by [Nathan Giusti](#).
- Baker demographic (age/occupation) data from the [R bakeoff package](#) and [Kaggle](#)
- Hand-collected 7-day UK viewer counts from [BARB](#), used to supplement missing viewer data for later seasons.

Giusti's Data

The technical challenge score is different from the signature and showstopper scores because it is a ranking rather than an absolute component score. This creates a scaling issue because the number of contestants changes across rounds as bakers are eliminated. For example, first place out of twelve bakers carries more information than first place out of three bakers, and third place out of three is the worst performance in that episode even though rank three would be strong early in the season.

For example,

- Round 1: rank 1 out of 12 is very strong.
- Round 9: rank 1 out of 3 is still strong, but less informative.
- Round 1: rank 6 out of 12 is middle.
- Round 9: rank 3 out of 3 is worst.

For this reason, I make another variable so that we need not treat the raw technical rank as directly comparable across rounds. I transform it into a within-episode relative rank,

$$\text{tech_scaled}_{ie} = 1 - 2 \frac{\text{tech}_{ie} - 1}{n_e - 1}.$$

where n_e is the number of bakers remaining in episode e . This maps the technical result to a stable relative scale: 1 is best, 0 is middle of the field, and -1 is worst.

To keep the model predictors on a common scale, I also rescale this to match the range of the signature and showstopper totals. Each signature and showstopper score combines flavor, appearance, and bake components, so the raw totals range from -3 to 3 . I then multiply the above by 3 to match